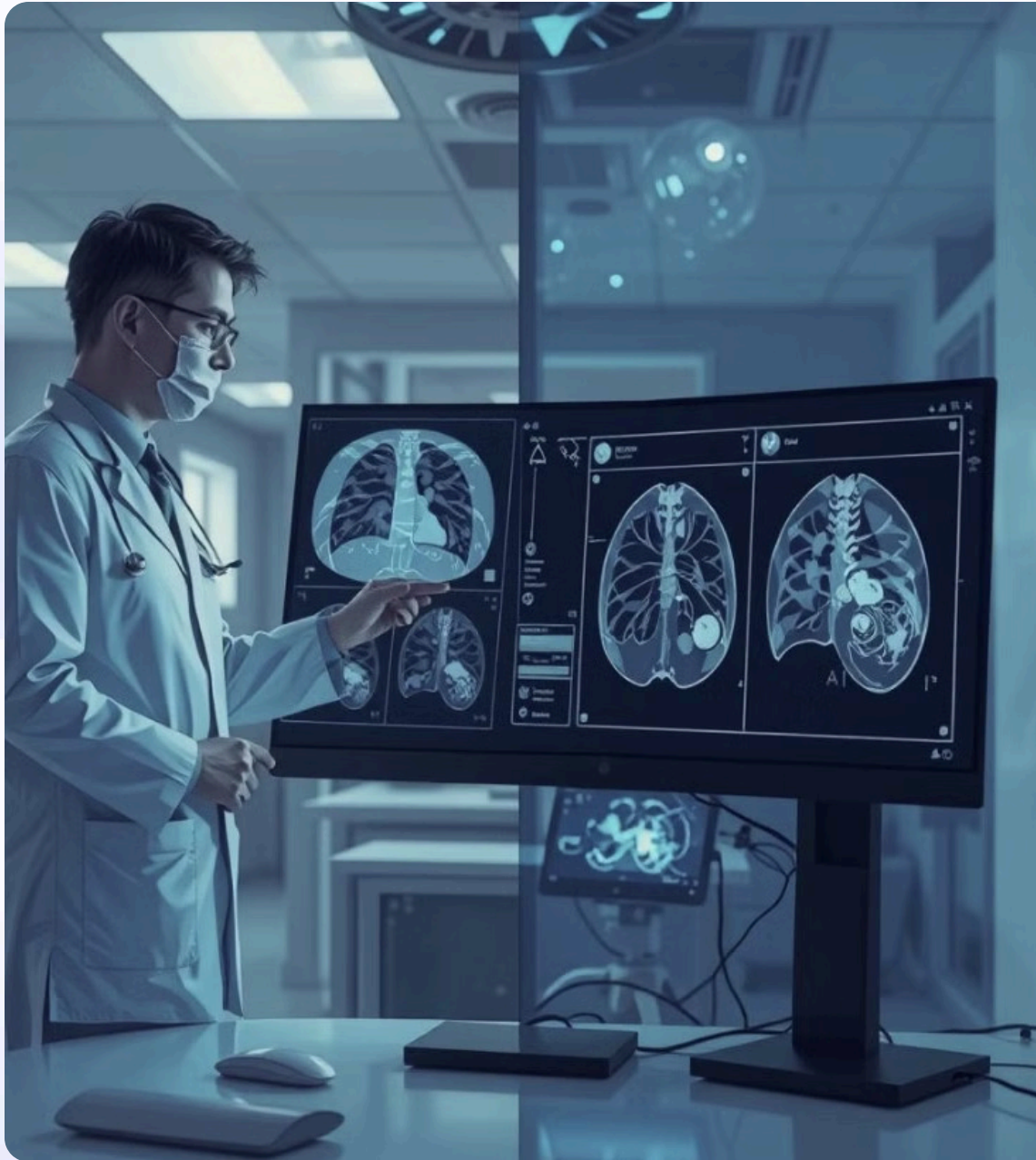
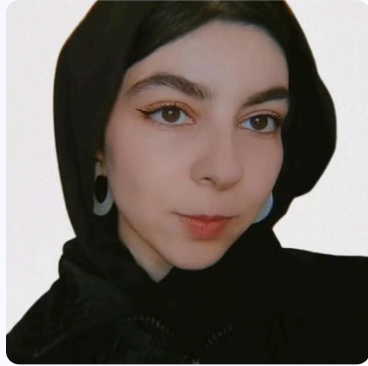




PulmoScan: AI for Lung Nodule Detection

Presented by : Data Wizards

Meet Our Team



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Data Scientist

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Email :

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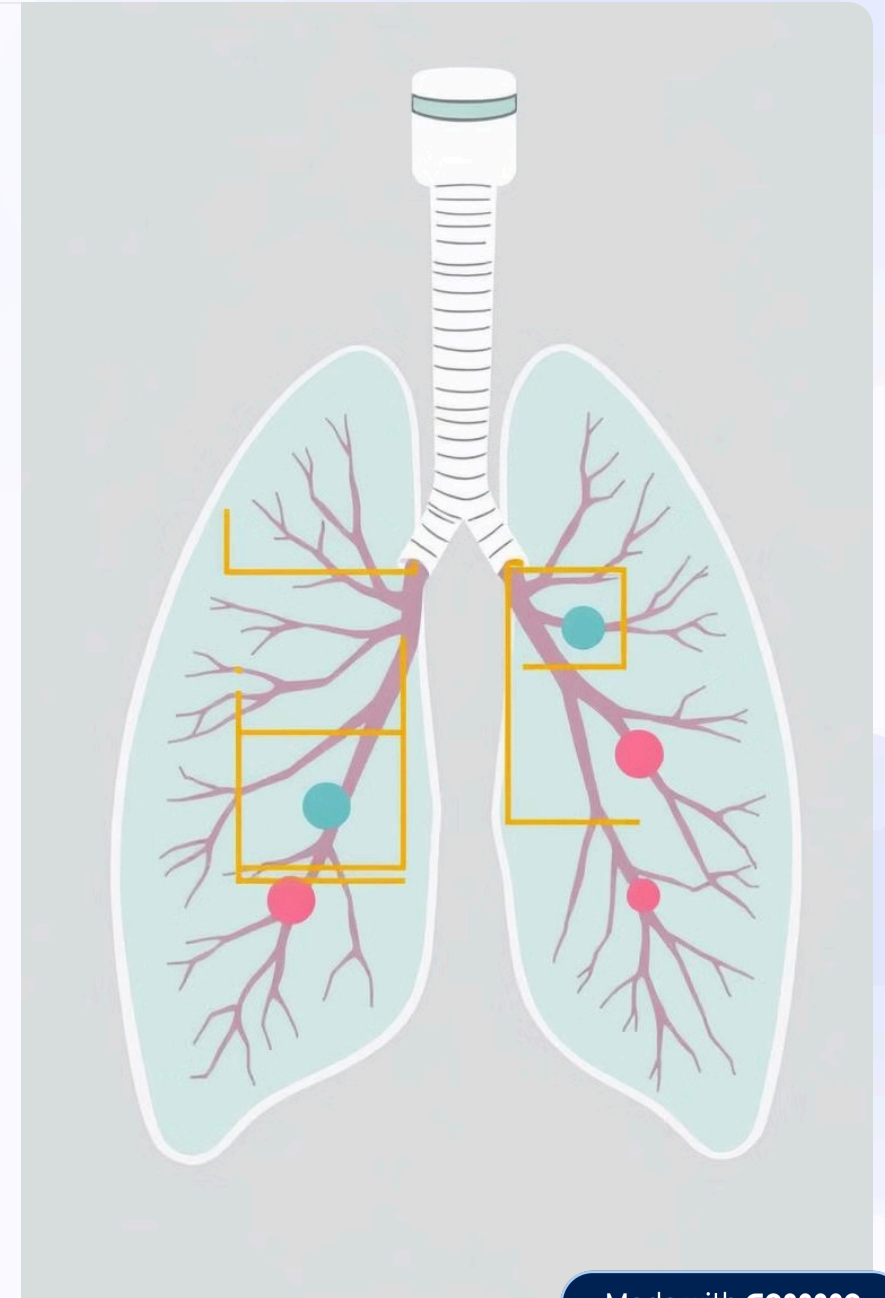
Table Of Content

- 1 Business Goals & Data Science Goals
- 2 Solution Architecture
- 3 Data Acquisition
- 4 Data Preparation
- 5 Data Visualisation
- 6 Steps To Do In The Future



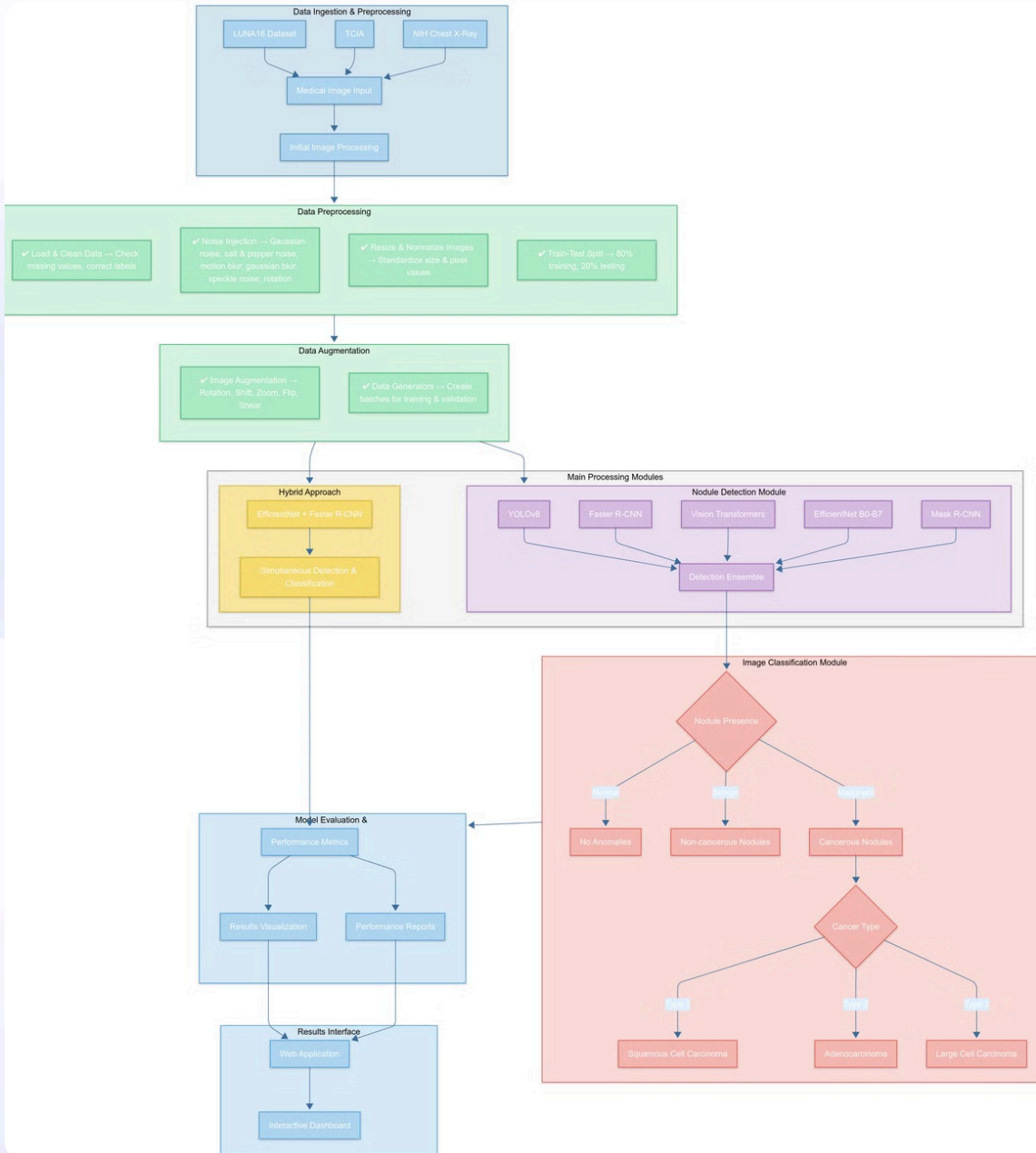
Introduction

- 1 Lung cancer is one of the deadliest diseases
- 2 Traditional diagnostics are slow and prone to errors
- 3 PulmoScan uses AI for high-accuracy lung nodule detection,
- 4 Revolutionizes lung nodule detection and classification



Business & Data Science Goals

Business Goals	Data Science Goals
Improve diagnostic accuracy by reducing human errors.	Reduce false positives and negatives using AI.
Reduce workload, allowing focus on complex cases.	Automate nodule detection and classification.
Make AI tools easy to use in hospitals and clinics.	Create a user-friendly system with visual outputs and reports.
Develop AI solutions for medical imaging and share findings through a scientific report.	Develop adaptable AI models with clear documentation for researchers and doctors.



Solution Architecture

[Click here](#)

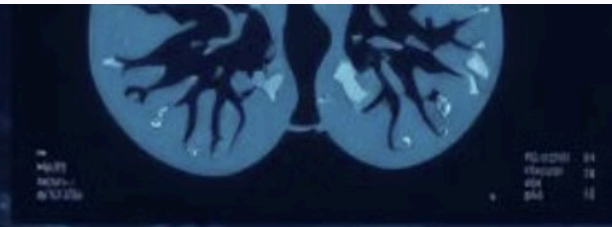


Data Acquisition: Building a Diverse Dataset

- 1 LUNA16 Dataset
- 2 TCIA (The Cancer Imaging Archive)
- 3 NIH Chest X-ray Dataset
- 4 The IQ-OTHNCCD lung cancer dataset

Dataset-Specific Data Science Goals

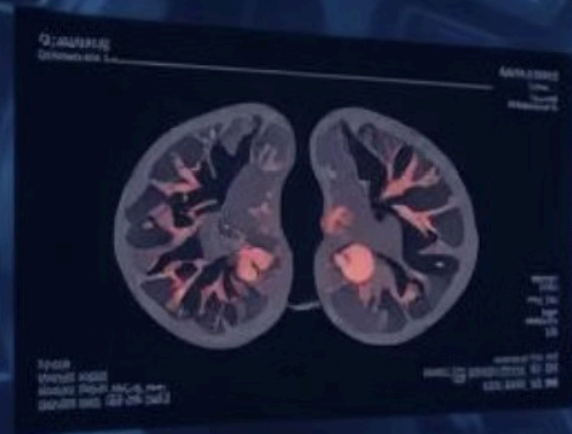
Dataset	Data Science Goal
LUNA16 Dataset & The IQ-OTHNCCD lung cancer dataset	Reduce false positives and false negatives using advanced AI-techniques
TCIA (The Cancer Imaging Archive)	Improve automated nodule detection and classification
NIH Chest X-ray Dataset	Create adaptable AI models with clear documentation for researchers and doctors.
Applies to All Datasets	Develop a user-friendly system with visual outputs and reports to help radiologists interpret results easily



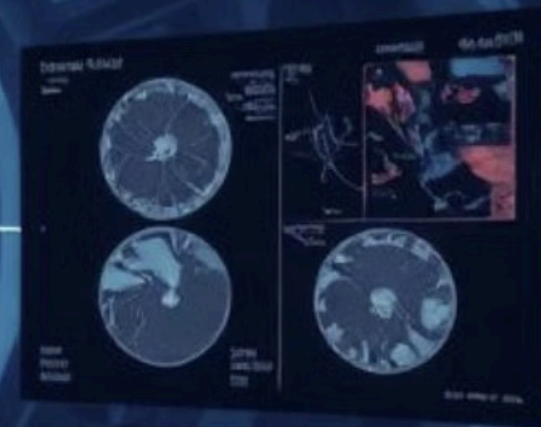
LUNA16, pri
LUI lung detections selecting



NICH X XRay
All-intastified cancer imaging
with 23d CT, MRI at X-ray



TCIA
AI classedfied imaging
cancer frel inlings



NIH Chest X ray
Radiory adattable, with
nilH-in the All.ouports

Data Preparation : Lunal6 Dataset

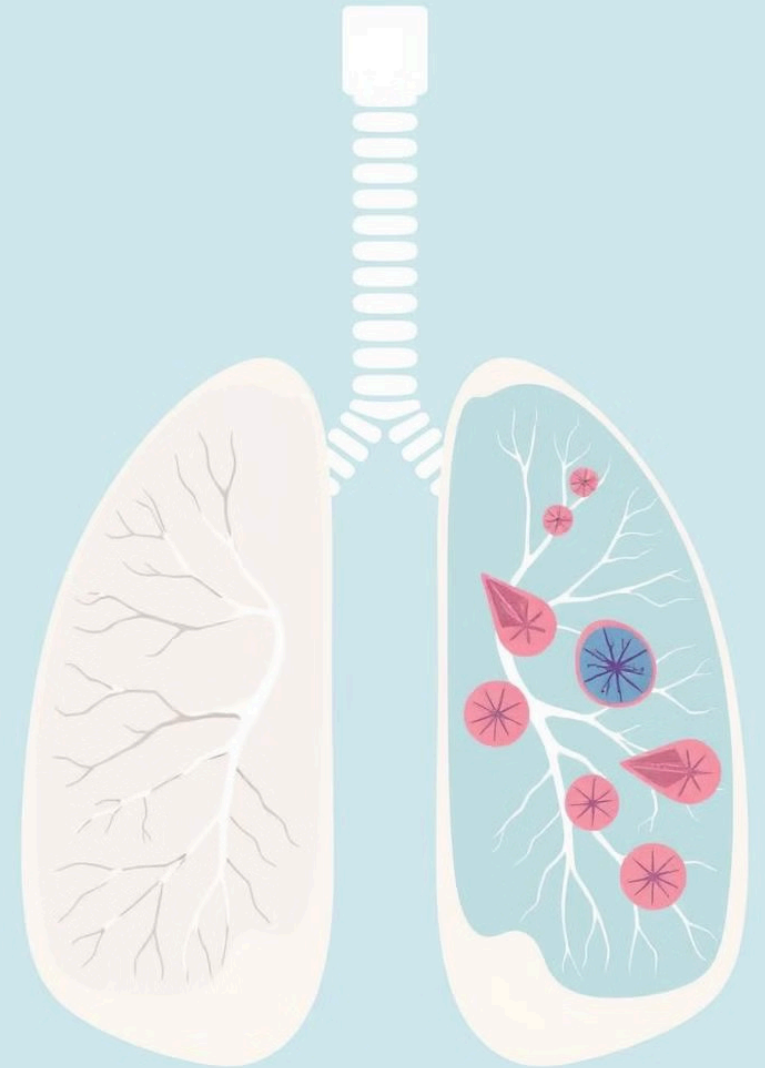
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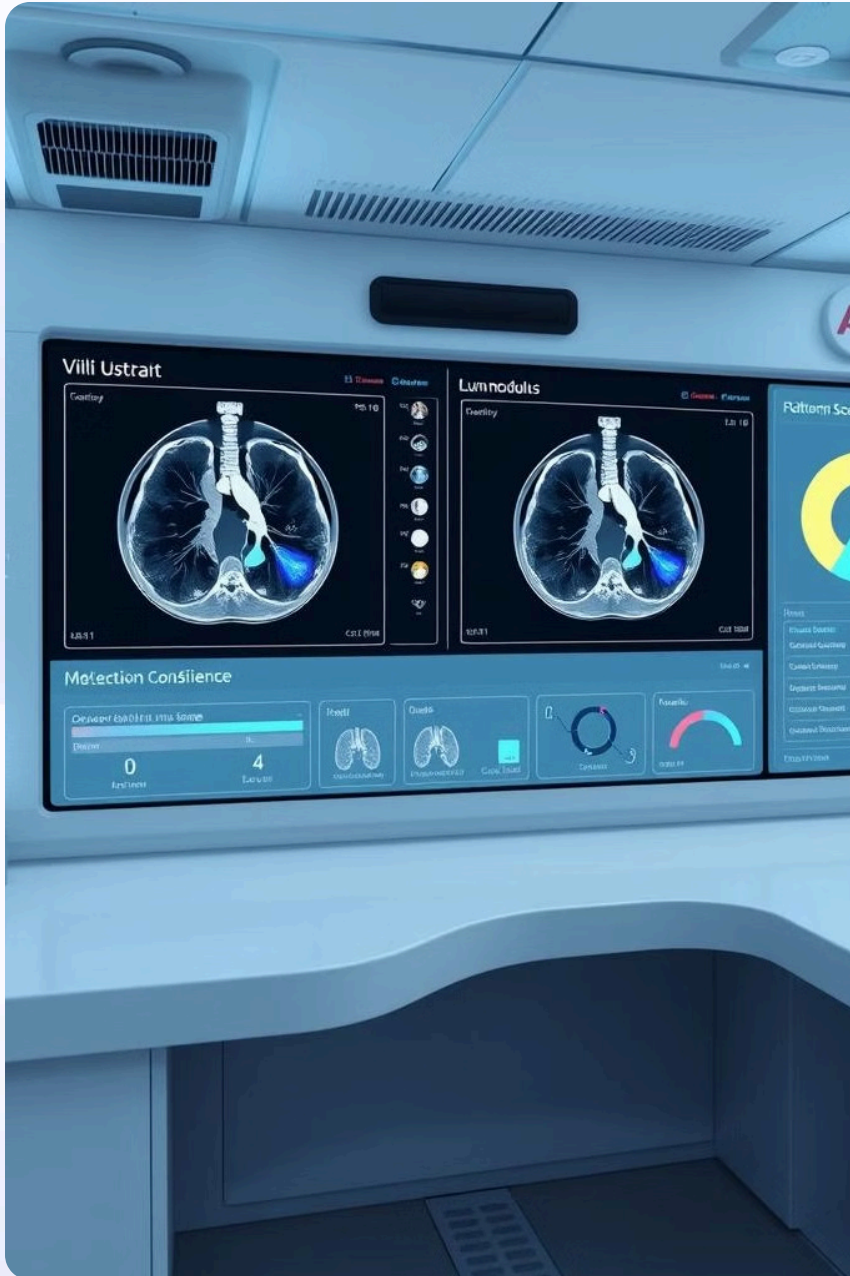
Gaussian Noise

2

Poisson Noise

Injecting **Gaussian** and **Poisson noise** helps our model generalize better, improving its accuracy in real-world medical imaging.





Data Preparation : TCIA (Cancer Imaging)

1

Motion Blur

2

Speckle Noise

Combining **Motion Blur** & **Speckle Noise** ensures AI models can accurately detect lung nodules in real-world medical scans.

Data Preparation : NIH Chest X-Ray Dataset

1

Contrast
Variation

2

Salt & Pepper
Noise

Using **Contrast Variation** and **Salt & Pepper Noise** together ensures AI models can accurately detect lung nodules even in poorly exposed or degraded X-rays.





Data Preparation : The IQ-OTHNCCD Dataset

1

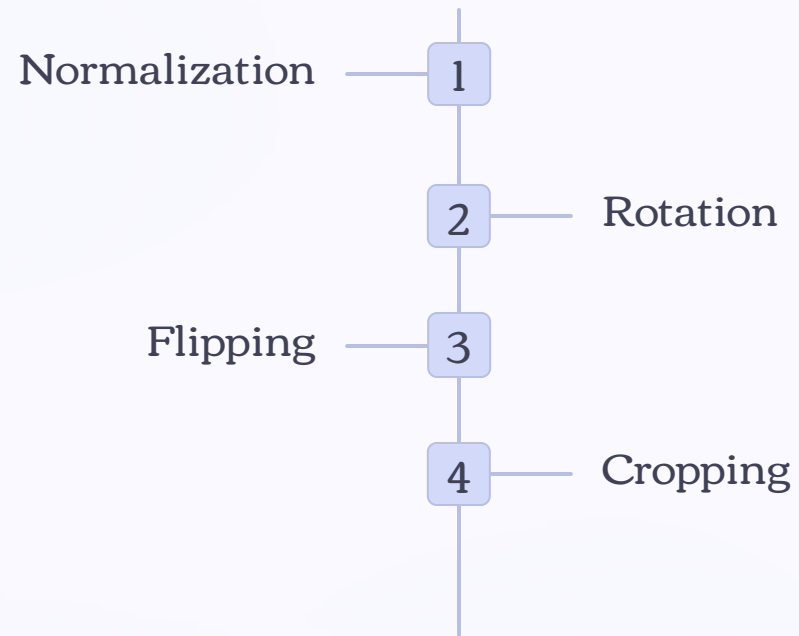
Gaussian Blur

2

Contrast
Variation

Gaussian blur and **Contrast Variation** boost AI reliability by smoothing details and simulating lighting changes for better lung nodule detection.

Data Preparation : Applied to all the datasets

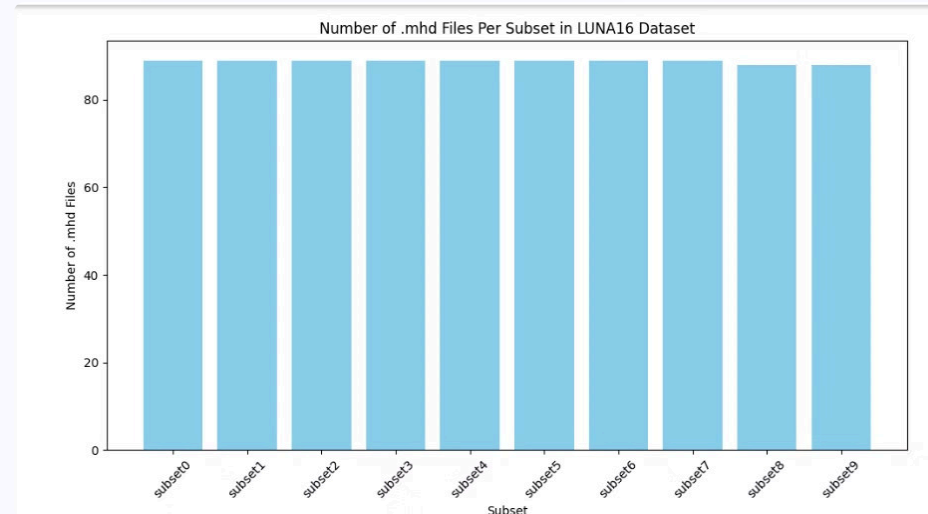


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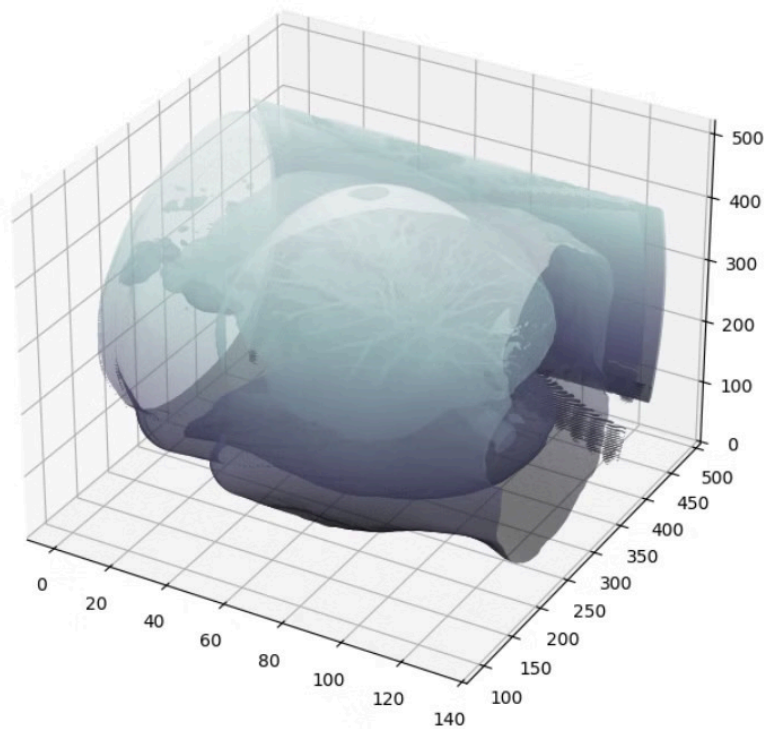


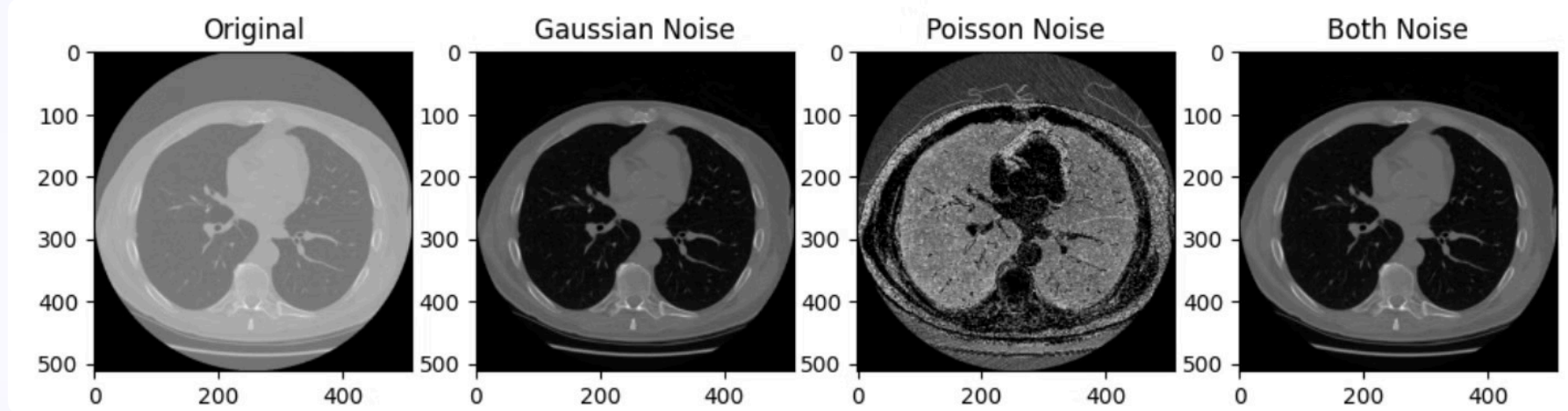
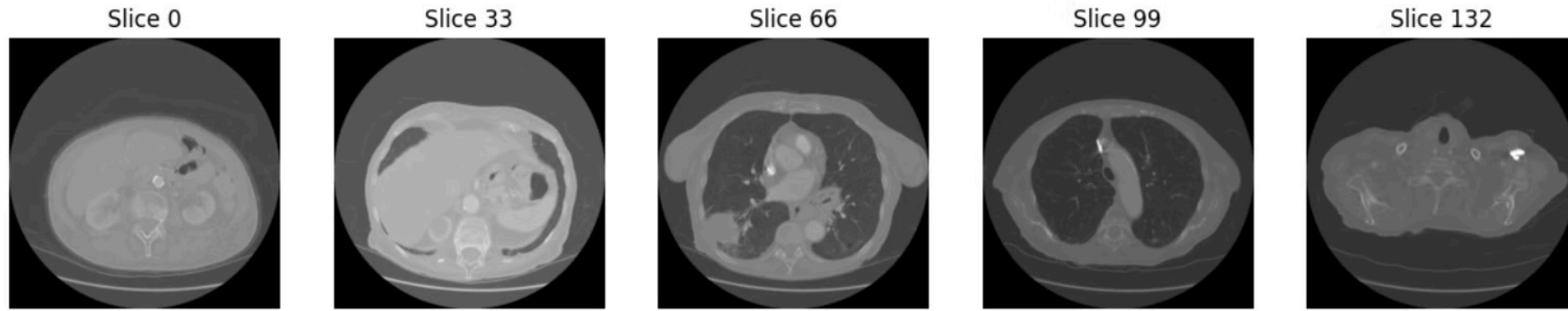
Luna16 Visualisation

- subset9
- subset8
- subset7
- subset6
- subset5
- subset3 (1)
- subset4
- subset2
- subset1
- subset0
- seg-lungs-LUNA16
- evaluationScript
- sampleSubmission
- candidates_V2
- candidates
- annotations



3D Visualization of Lung





Annotations loaded. Total nodules: 1186

	seriesuid	coordX	coordY
0	1.3.6.1.4.1.14519.5.2.1.6279.6001.100225287222...	-128.699421	-175.319272
1	1.3.6.1.4.1.14519.5.2.1.6279.6001.100225287222...	103.783651	-211.925149
2	1.3.6.1.4.1.14519.5.2.1.6279.6001.100398138793...	69.639017	-140.944586
3	1.3.6.1.4.1.14519.5.2.1.6279.6001.100621383016...	-24.013824	192.102405
4	1.3.6.1.4.1.14519.5.2.1.6279.6001.100621383016...	2.441547	172.464881

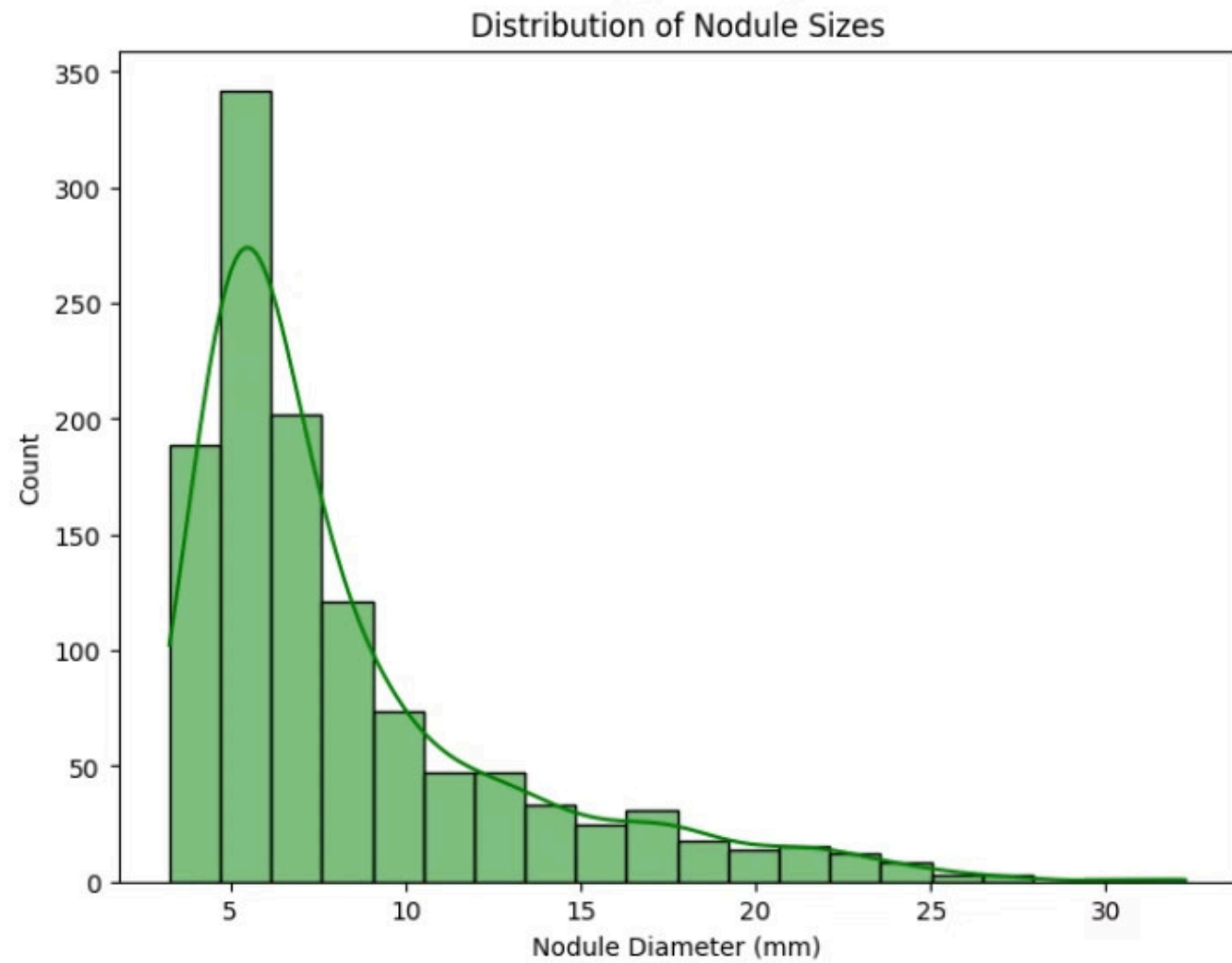
	coordZ	diameter_mm
0	-298.387506	5.651471
1	-227.121250	4.224708
2	876.374496	5.786348
3	-391.081276	8.143262
4	-405.493732	18.545150

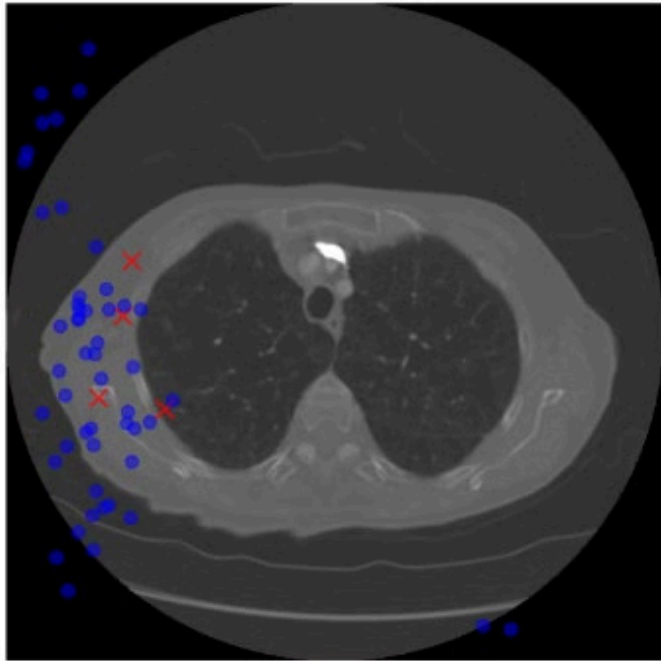
14

Candidates loaded. Total candidate locations: 551065

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1	1.3.6.1.4.1.14519.5.2.1.6279.6001.100225287222...	53.21	-244.41	-245.17
2	1.3.6.1.4.1.14519.5.2.1.6279.6001.100225287222...	103.66	-121.80	-286.62
3	1.3.6.1.4.1.14519.5.2.1.6279.6001.100225287222...	-33.66	-72.75	-308.41
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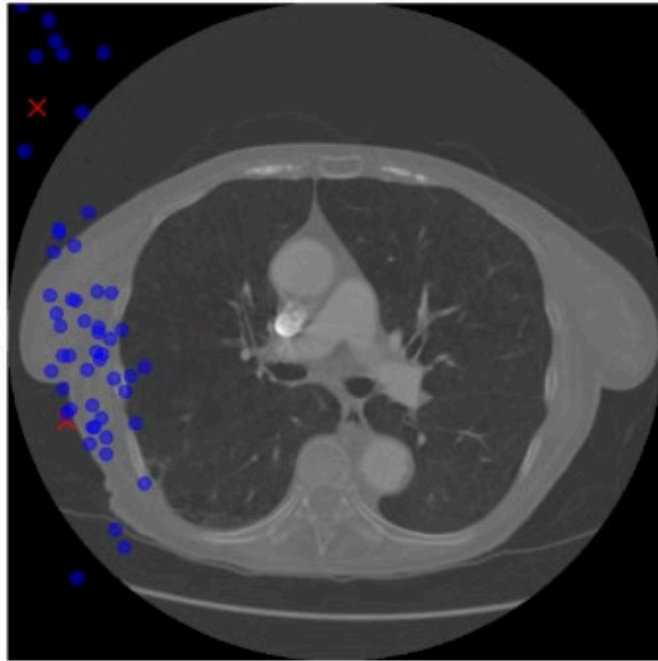
	class
0	0
1	0
2	0
3	0
4	0





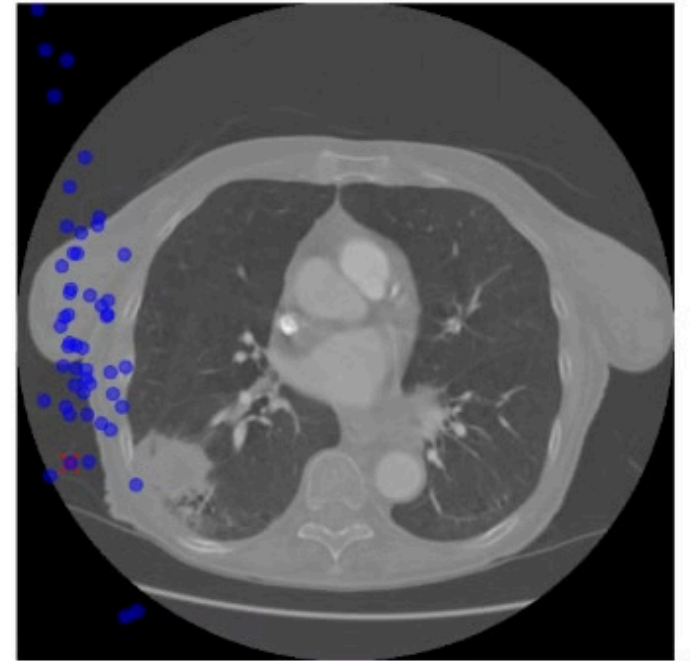
X Nodule (ID: 424)
● Candidate

Slice 89



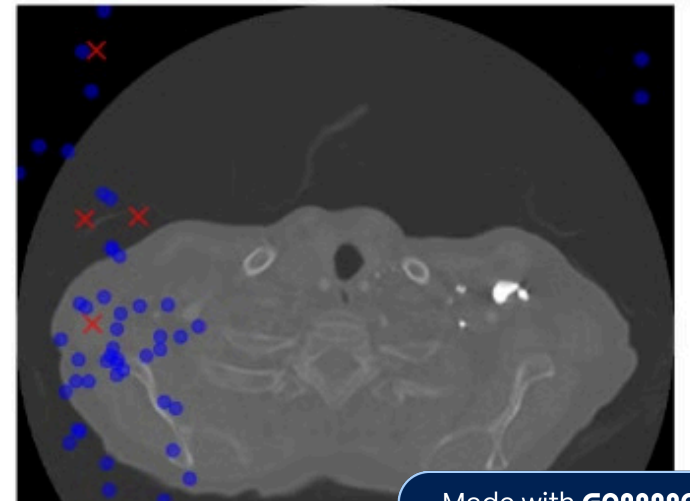
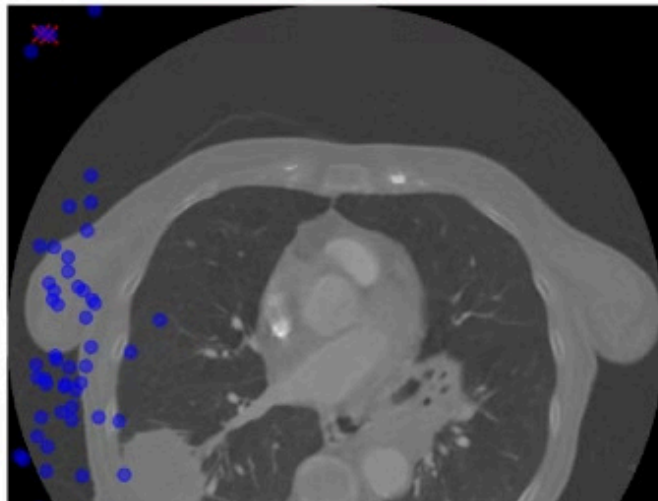
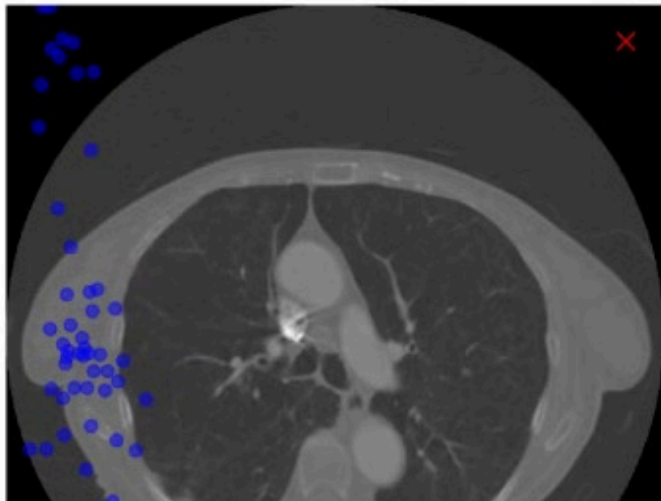
X Nodule (ID: 152)
● Candidate

Slice 64

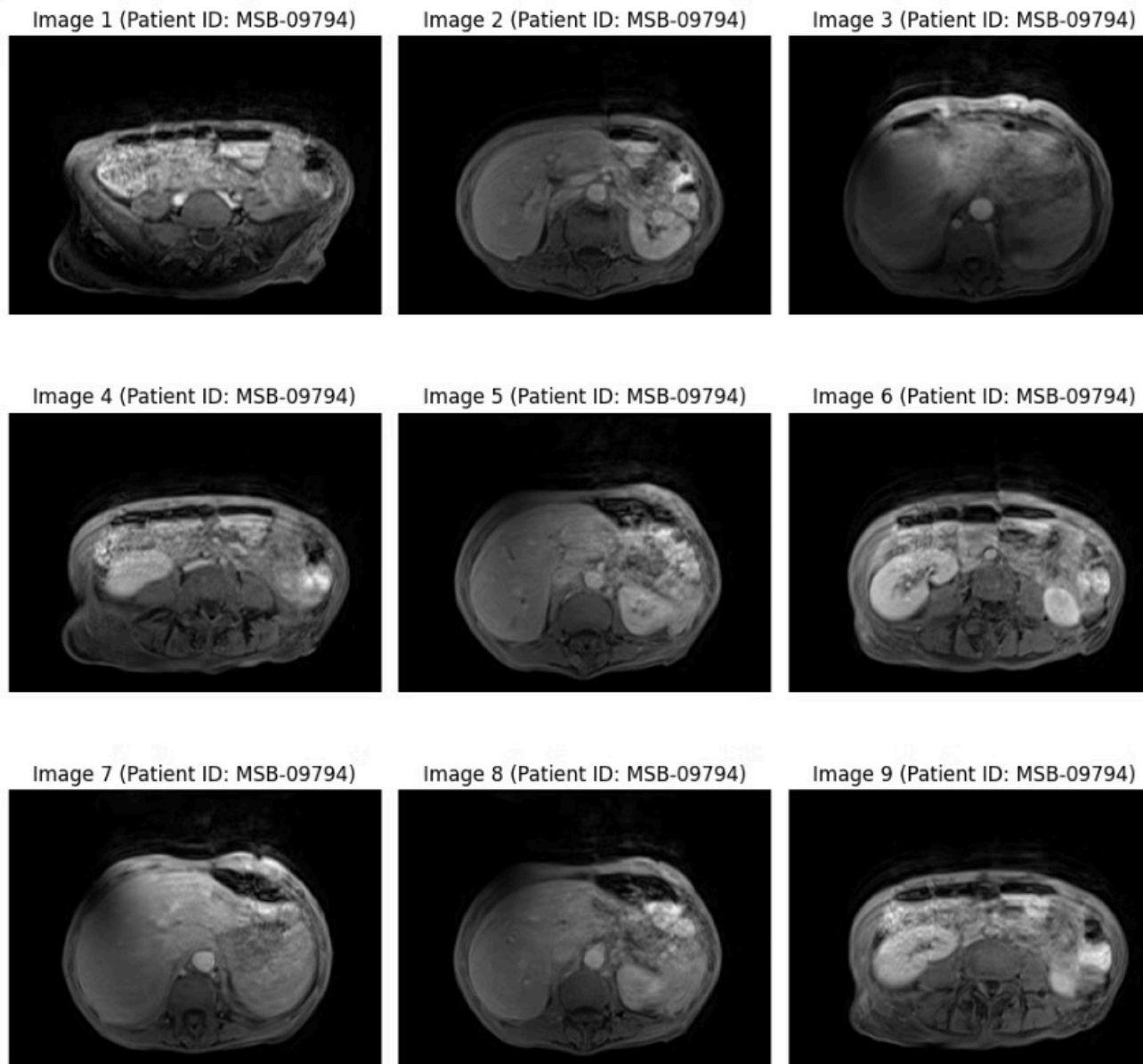


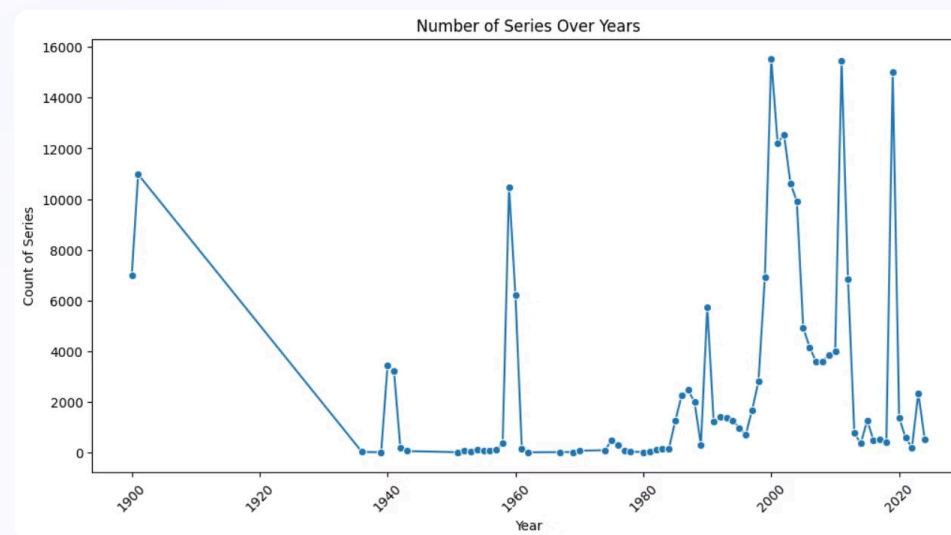
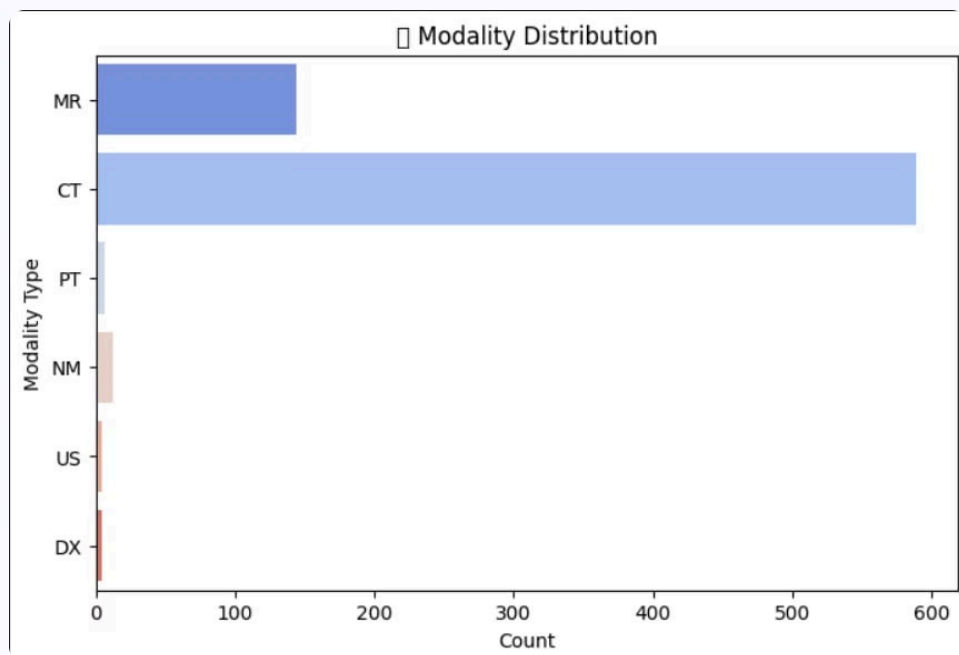
X Nodule (ID: 325)
● Candidate

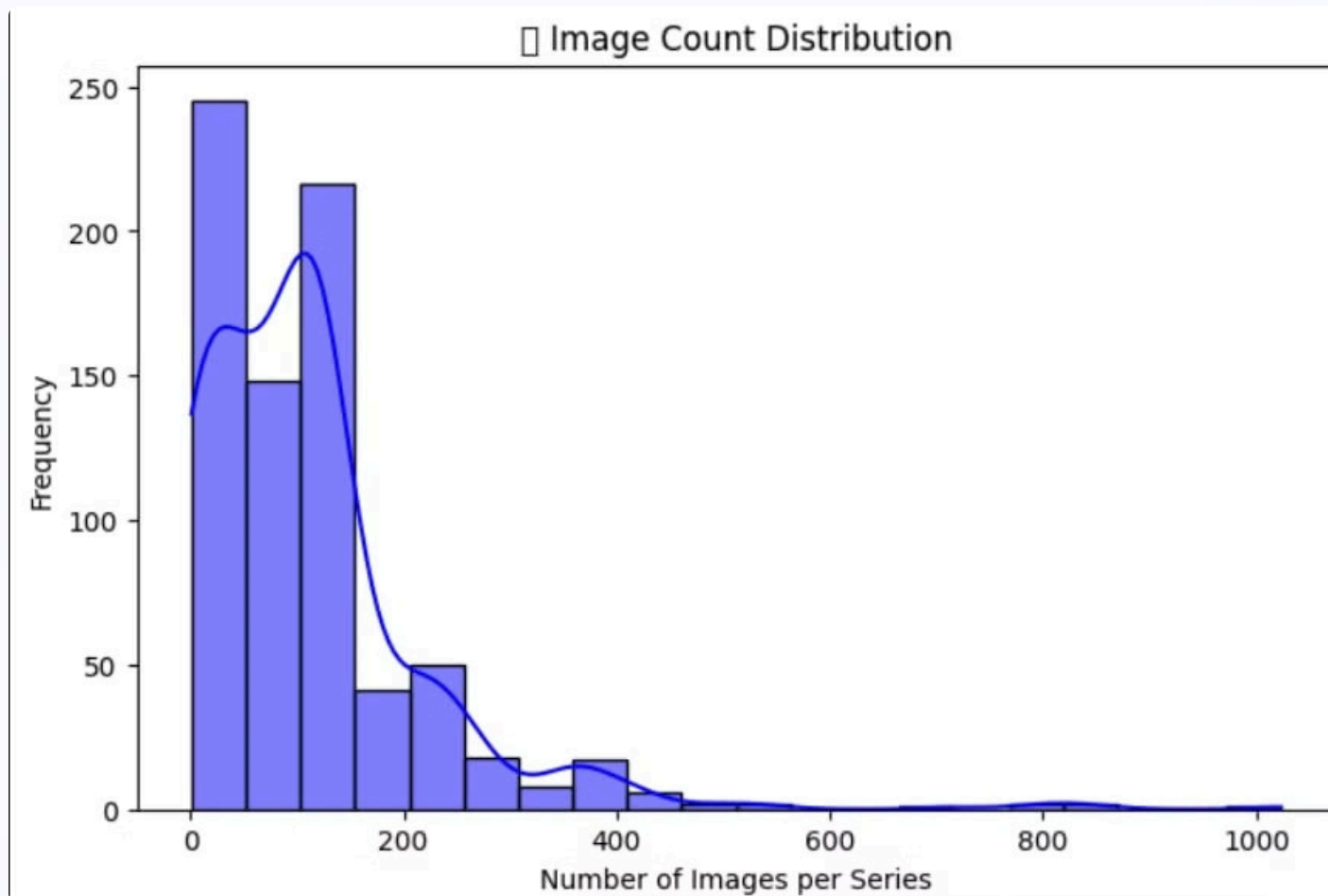
Slice 128

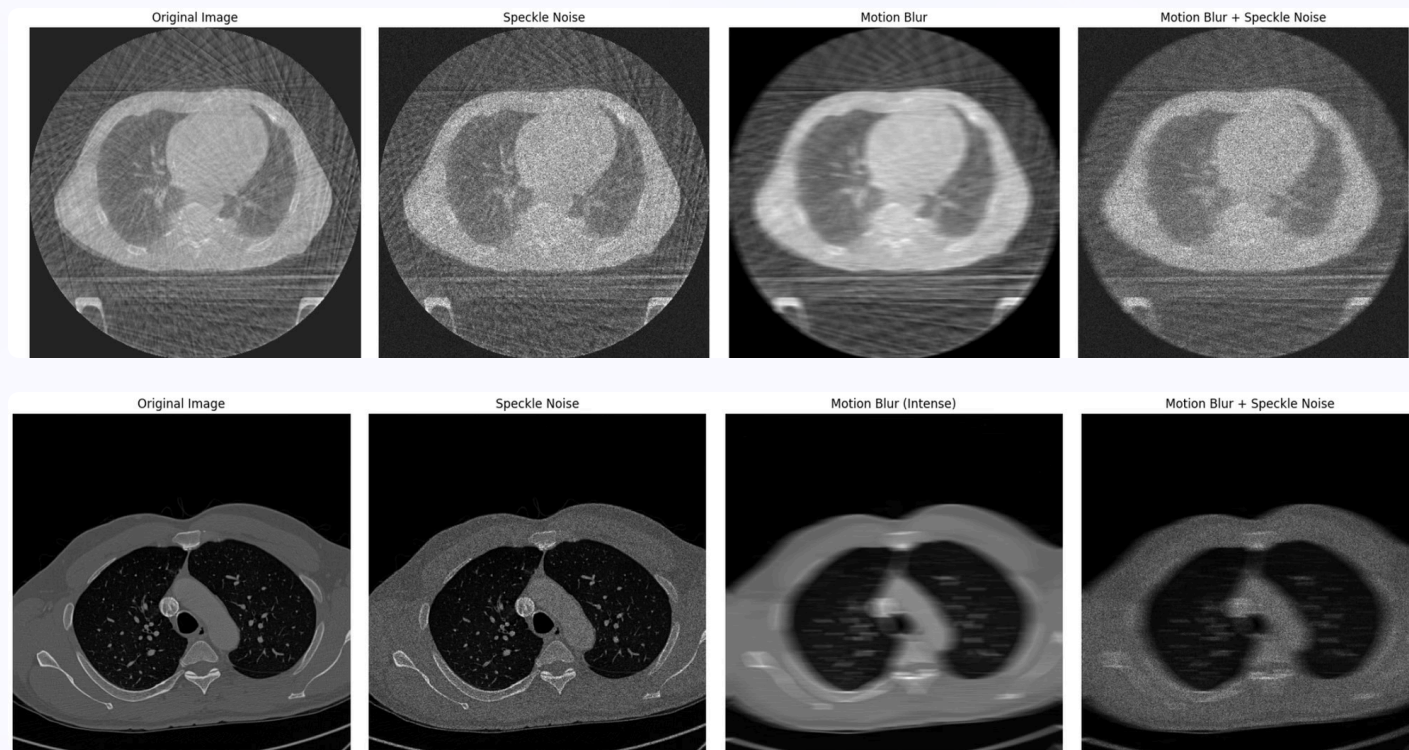


TCIA (Cancer Imaging) Visualisation

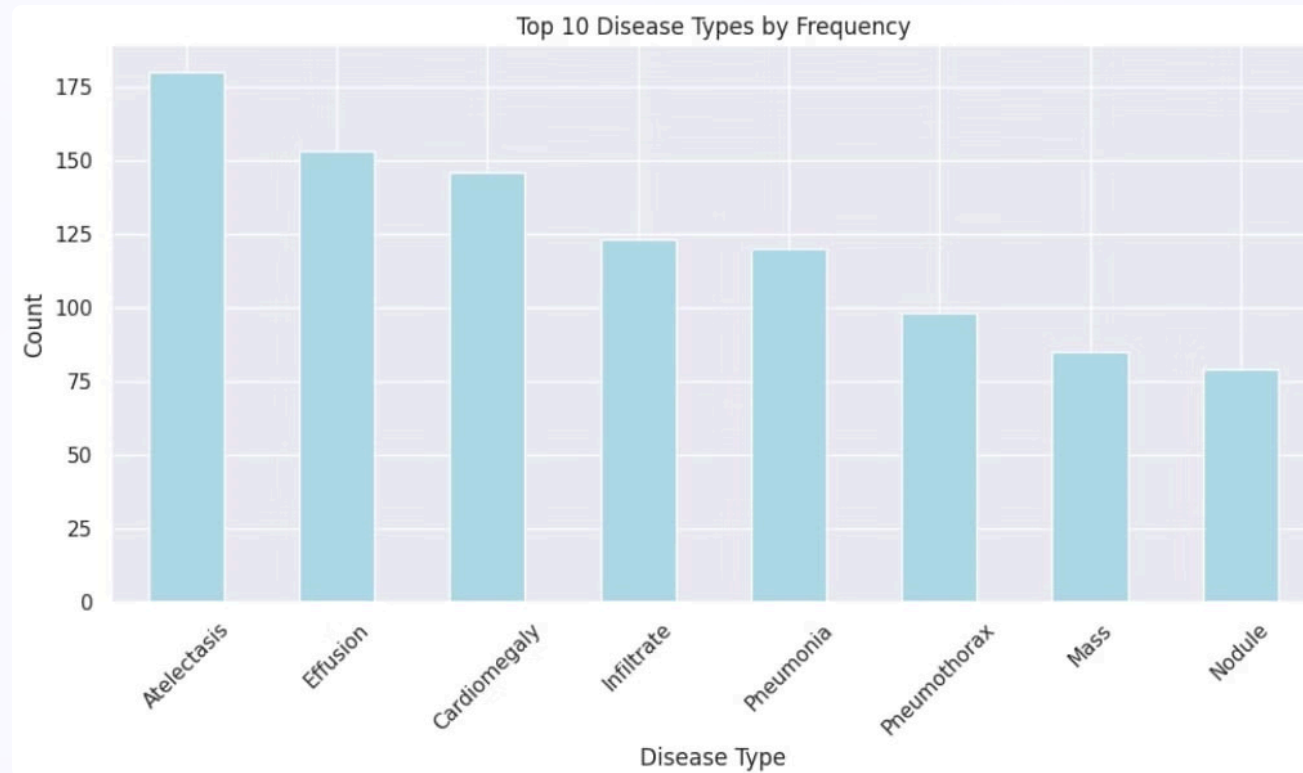


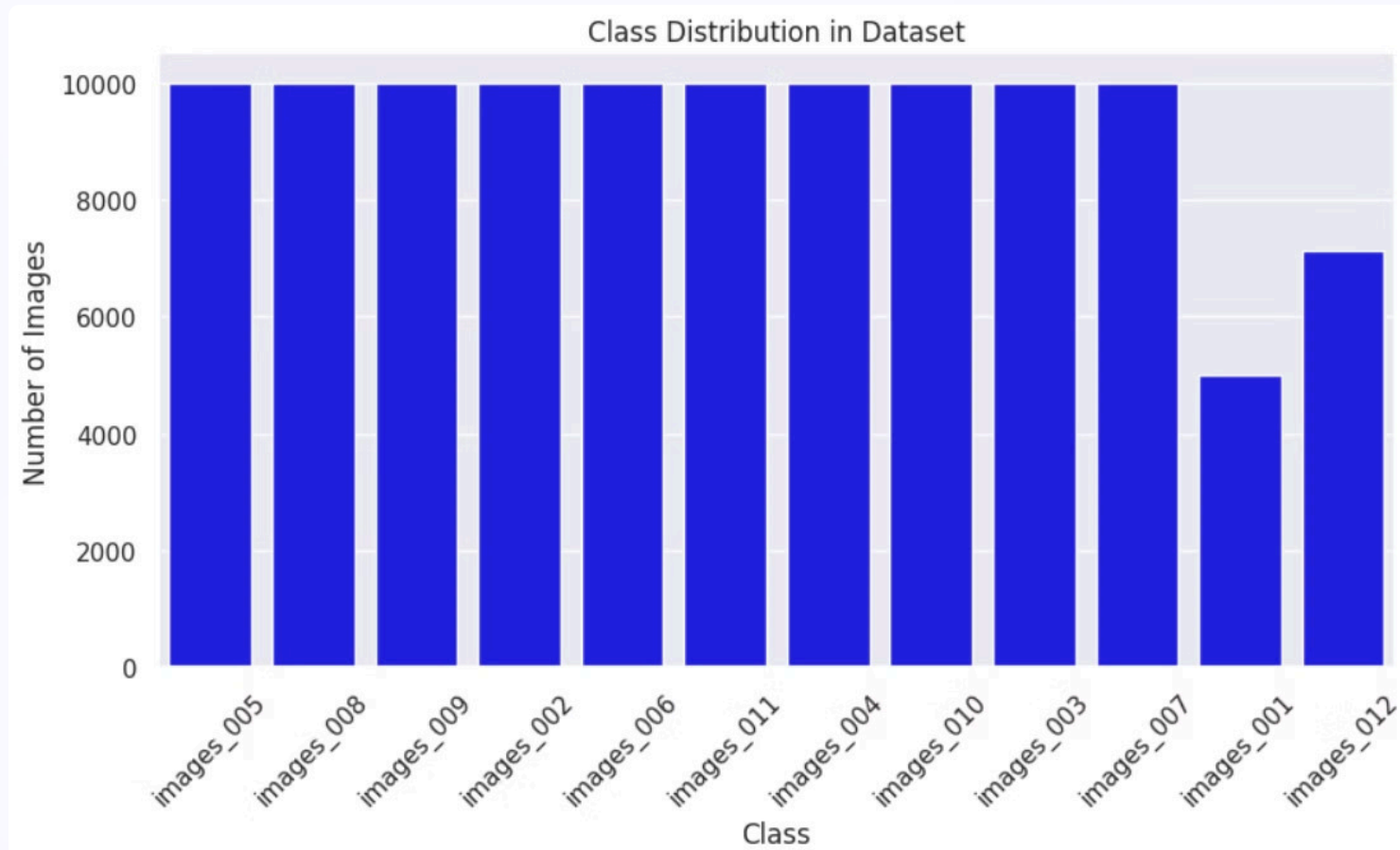


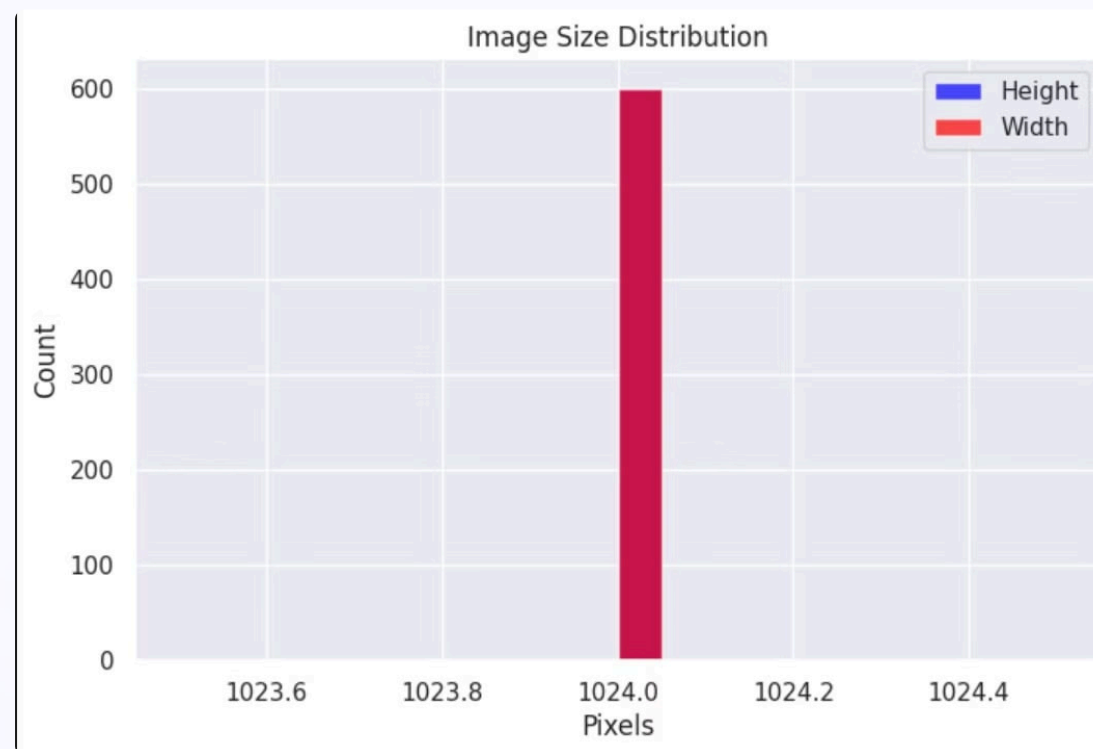
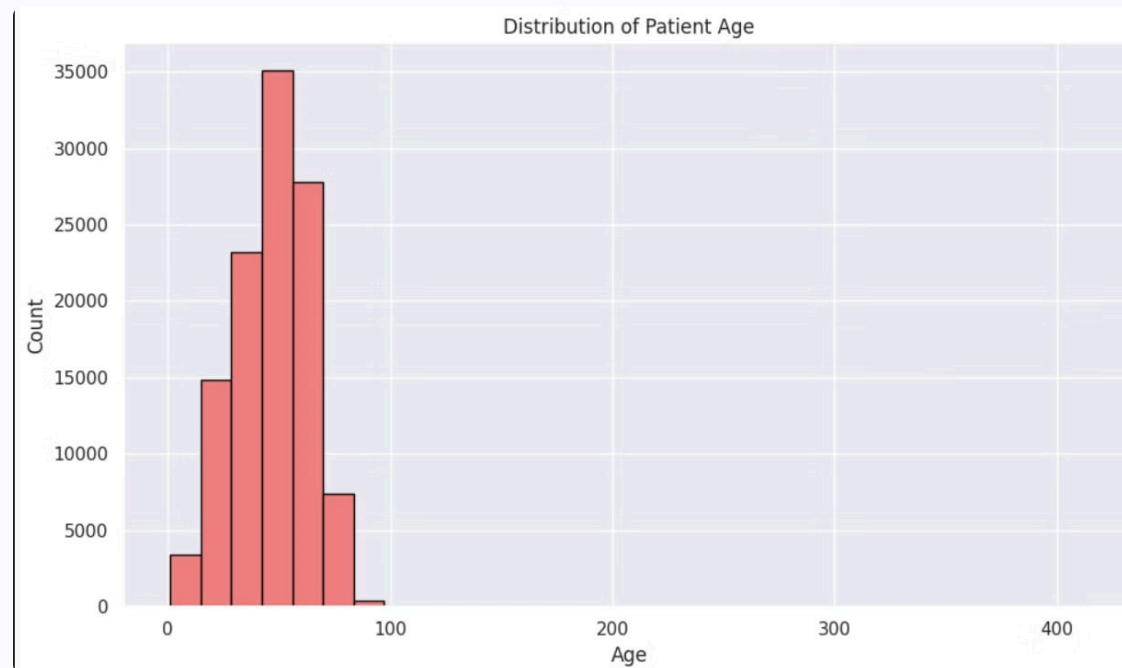


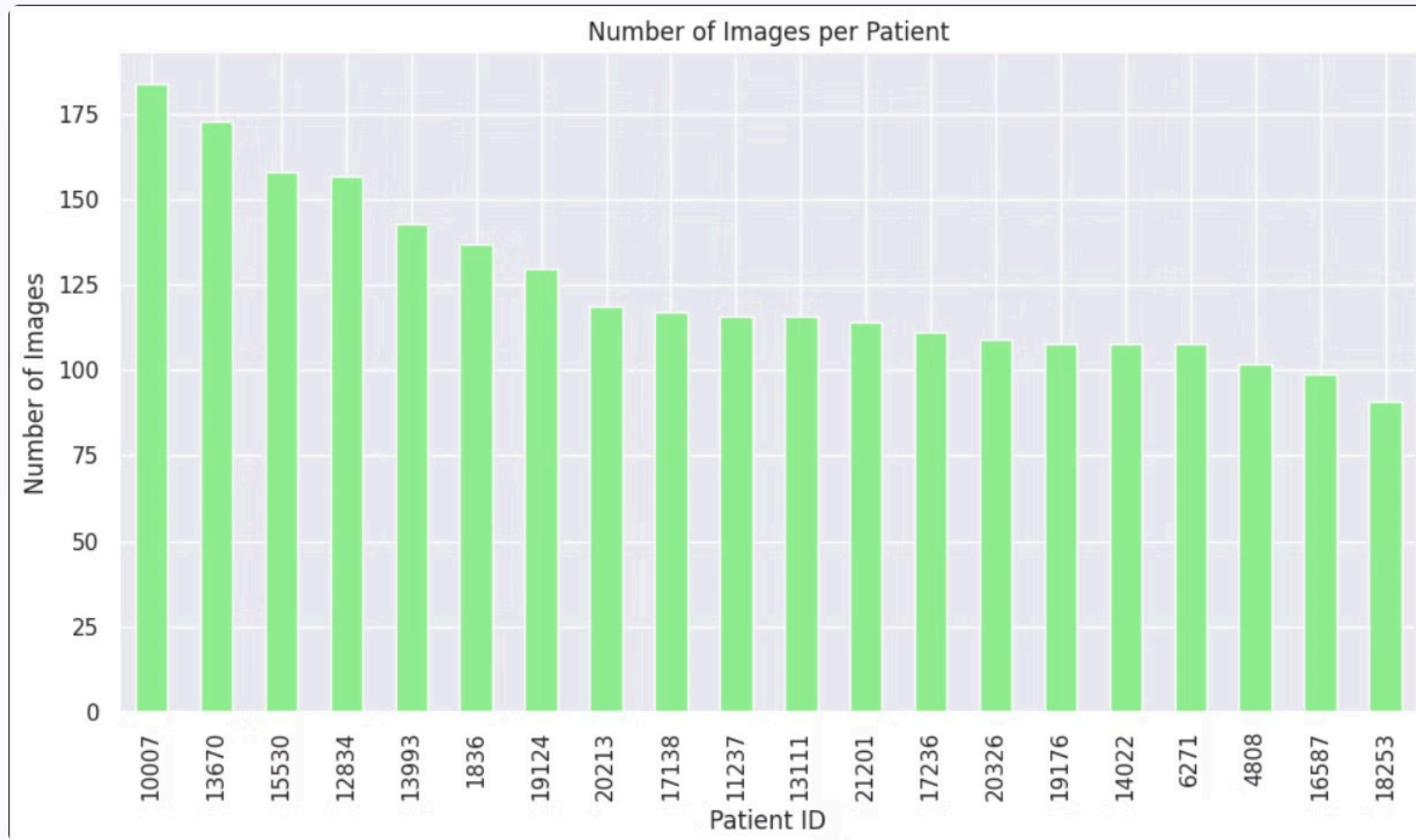


NIH Chest X-Ray Visualisation









Original 00020703_006.png



Processed 00020703_006.png



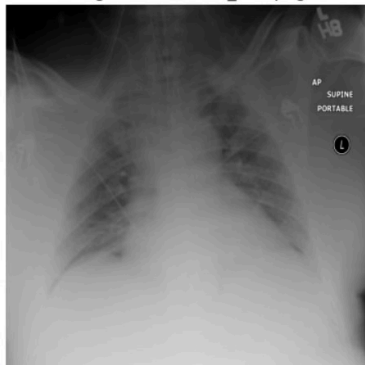
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Processed 00012640_002.png



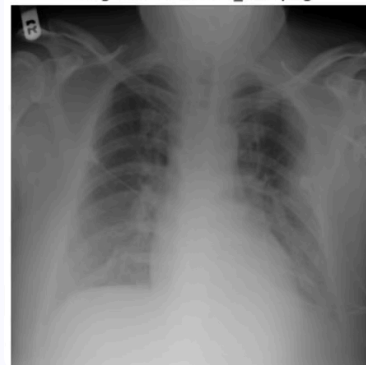
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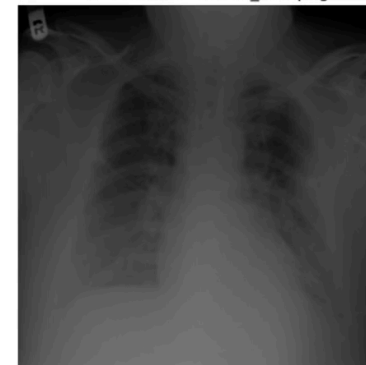
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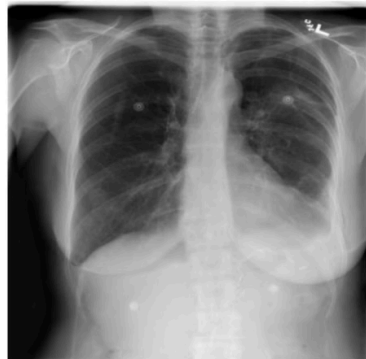
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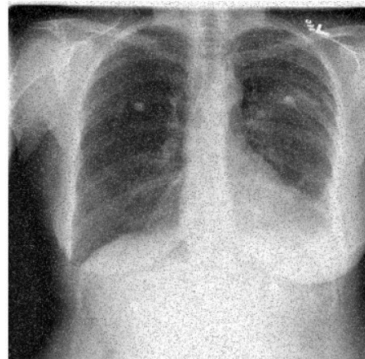
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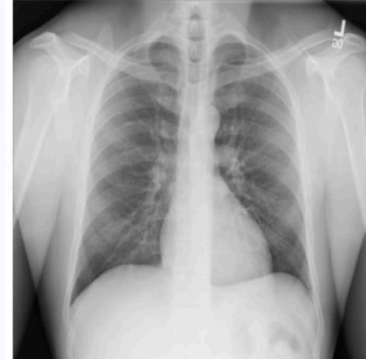
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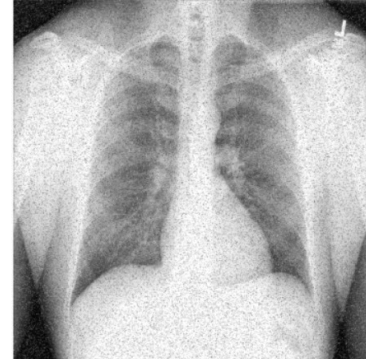
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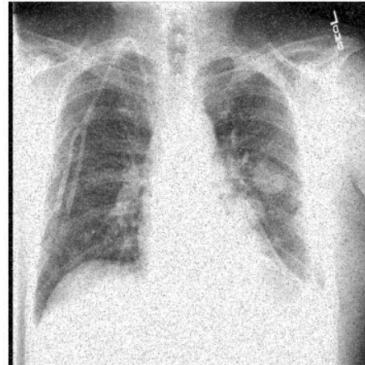
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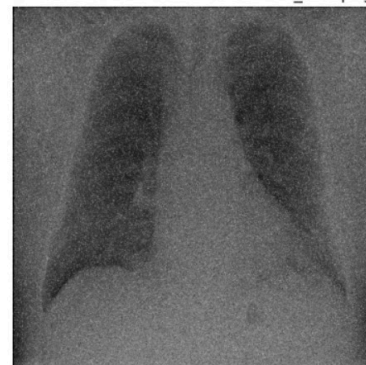
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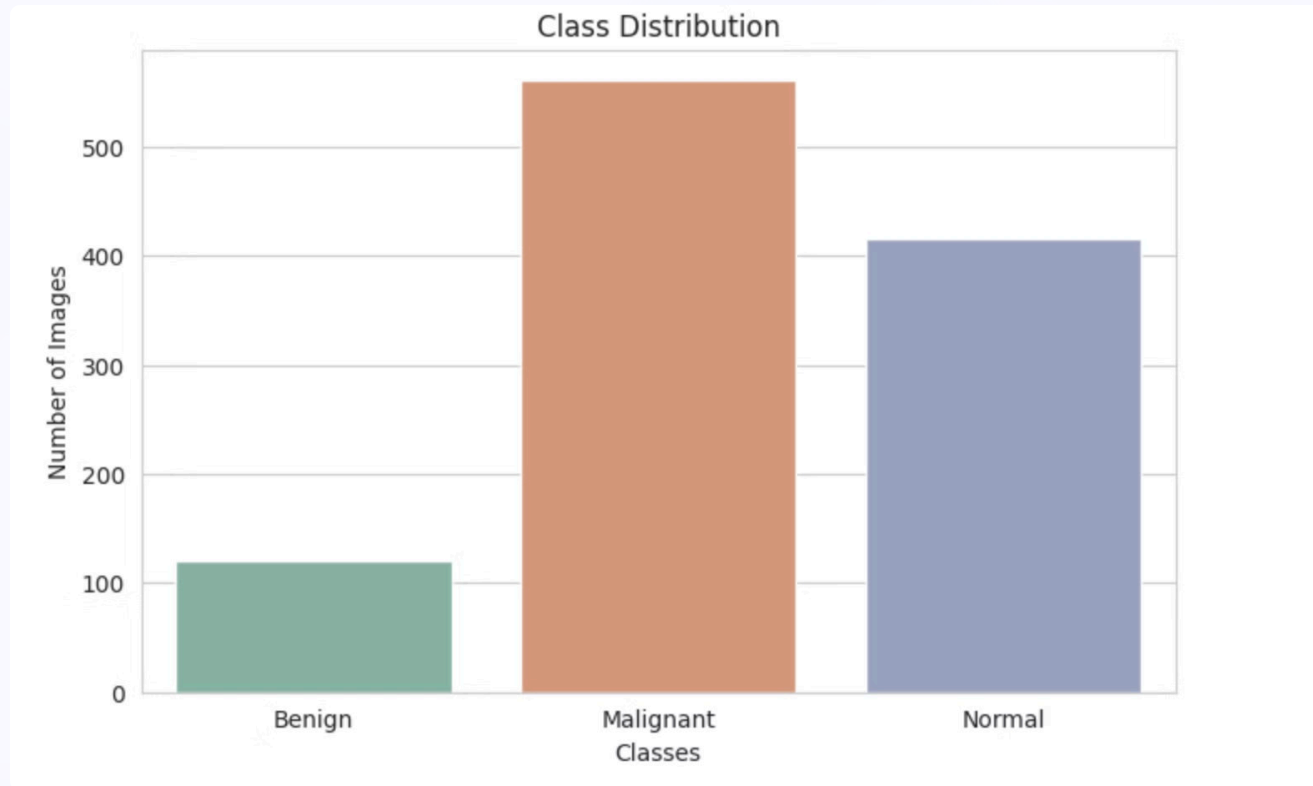
Original 00028766_000.png

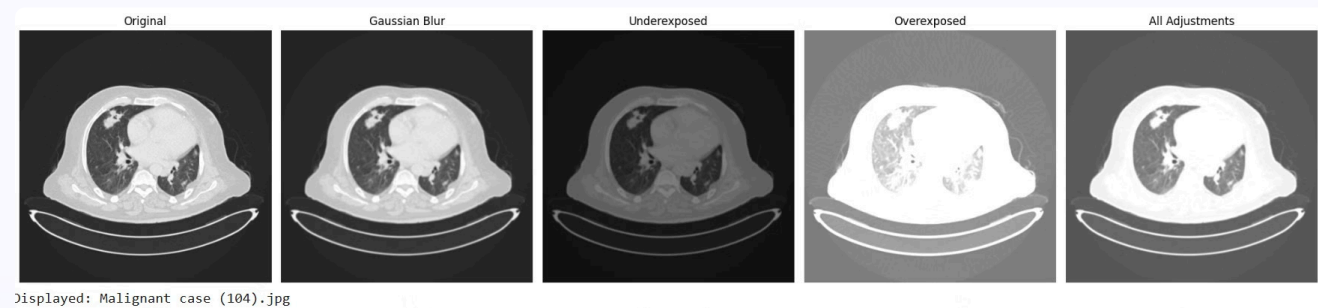
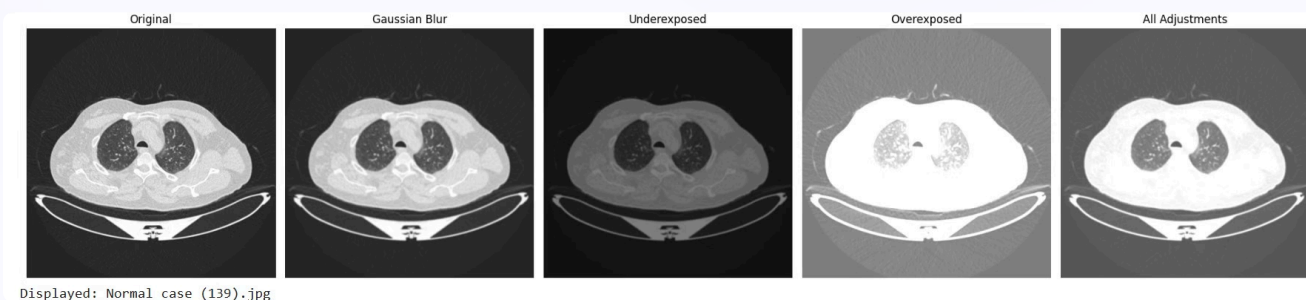
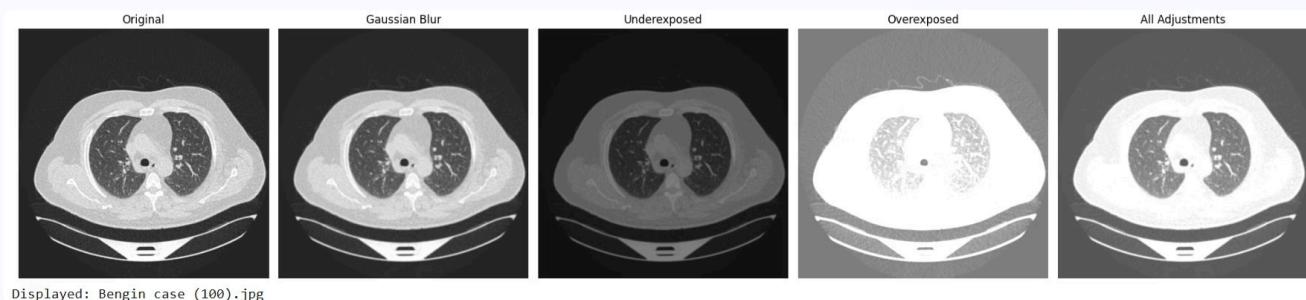


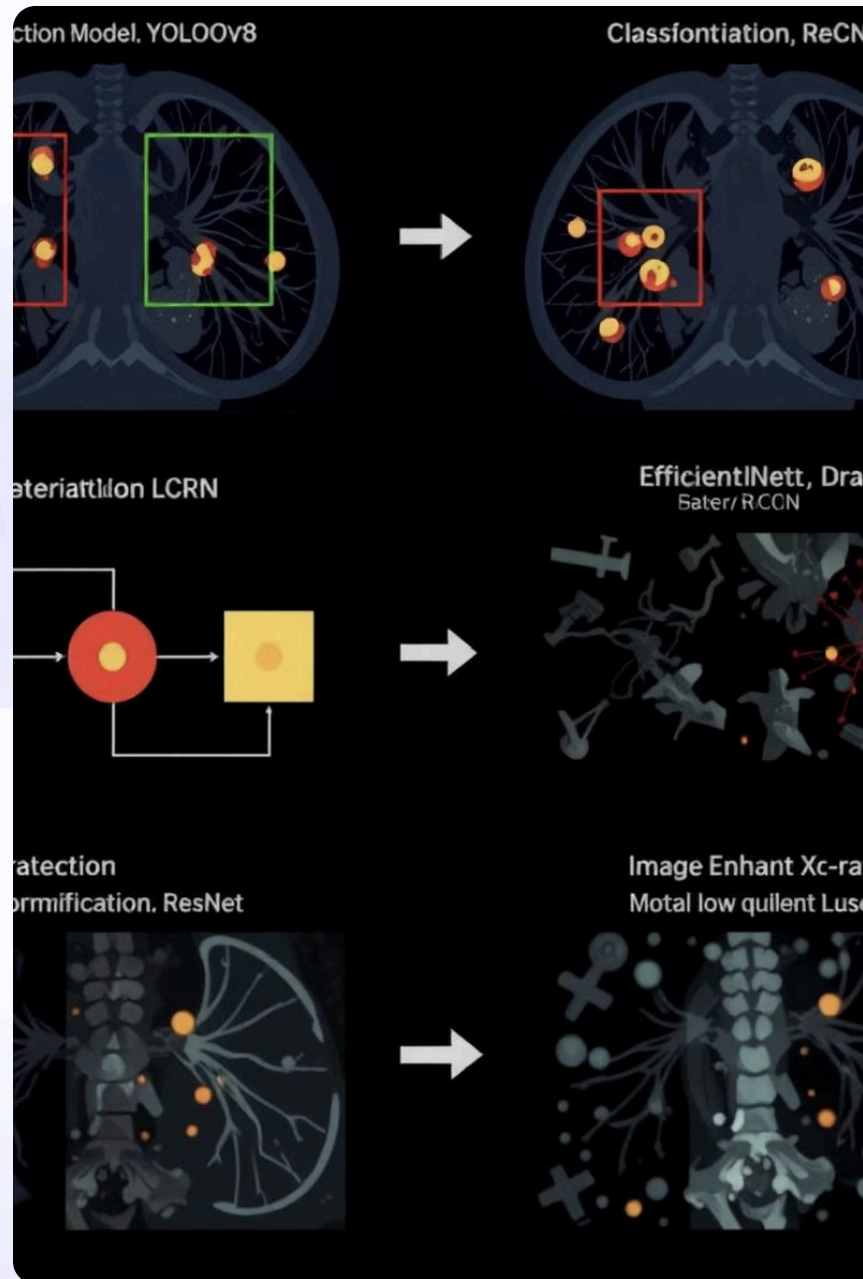
Processed with Noise 00028766_000.png



The IQ-OTHNCCD lung cancer Visualisation







Steps To Do In The Future: AI Model Training

1

Detection Models
(YOLOv8, Faster R-CNN)
to locate lung nodules.

2

Classification
Models
(EfficientNet, DenseNet)
to categorize nodules .

3

Hybrid Models
Combining detection & classification for efficiency (e.g.,
EfficientNet + Faster R-CNN).

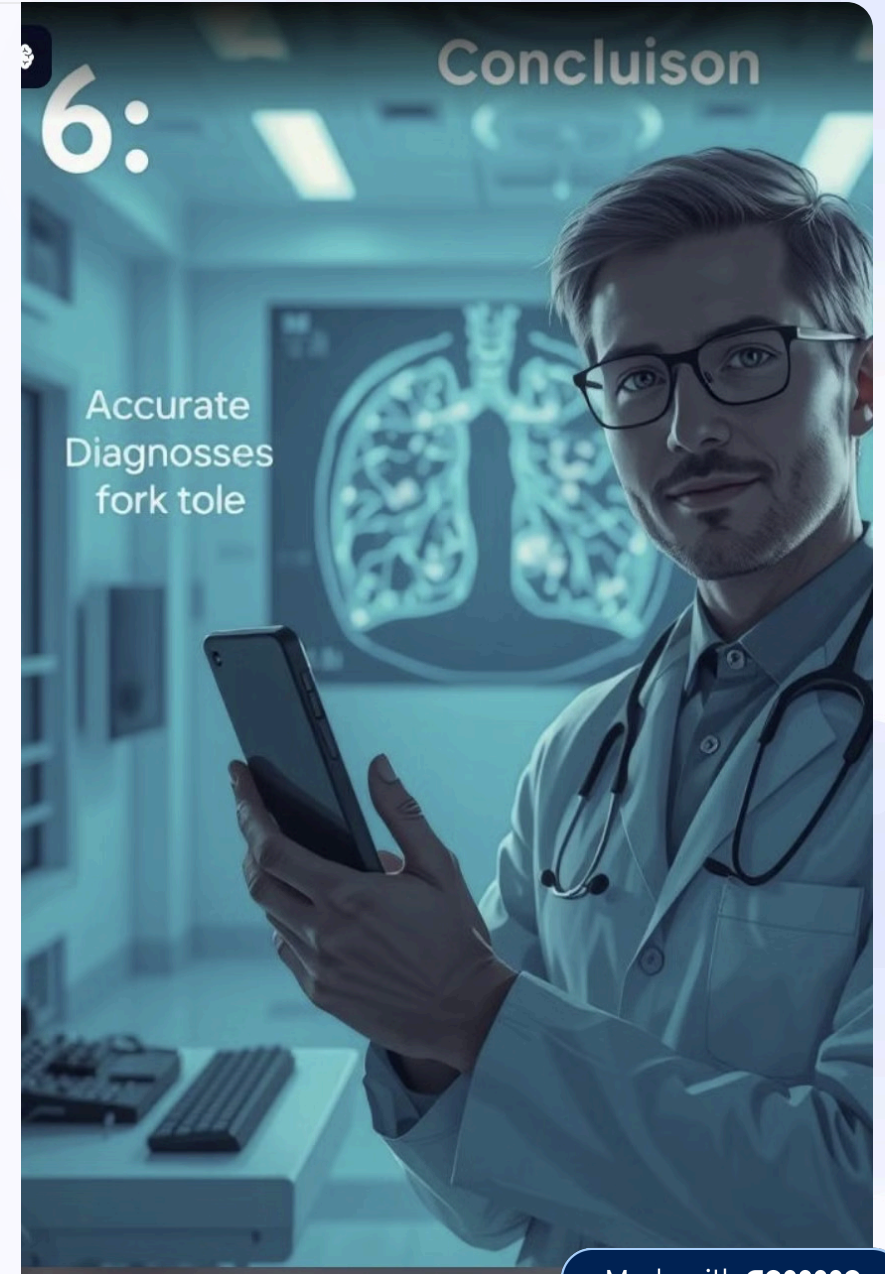
Steps To Do In The Future : Model Evaluation & Performance Metrics

- 1 — For Detection Models
Mean Average Precision .
- 2 — For Classification Models
Accuracy, Precision, Recall, F1-Score.
- 3 — Detection & Classification Models
ROC-AUC Score (to measure diagnostic reliability).



Conclusion

- 1 Revolutionizes Detection
- 2 Empowers Radiologists
- 3 Improves patient outcomes



Conclusion

6:

Accurate
Diagnoses
for the

Thank you for your
attention!

